Shared Data Reference

Civic Operations . Group 23 . the names every component uses

Purpose of this document

One place for the names every component shares. The intake bot turns a citizen's message into a request. Every component after it reads that request. If two components use different words for the same thing, a request can be routed wrongly or dropped without anyone noticing. This document gathers those words in one place: the request types, the field names and their allowed values, and the rules we agreed. It is meant as a helping hand, so everyone can build with confidence and link up easily.

- **How to read it.** Pages 1 to 3 once. Then your own page, which lists what your components receive, what they hand on, and the points we still need to discuss.
- **What it does not cover.** How any component works inside. Each owner designs their own component, and this document leaves that entirely to them.
- **Names, not numbers.** Components are named rather than numbered, because the numbers differ between documents. The list and owners come from the 28 September presentation. Admin questions was assigned to Nabeel on 14 October.
- **If something looks wrong.** The request types come from `data/reference/request_catalog.csv`, so that file is the reference. Please raise any change with the group, and the file and this document will be updated together.

How a request flows

Intake bot > Emergency check > Agency and skill routing > Duplicate detection > External factors > Credibility > Fake request detection > Resolution time > Urgency > Prioritisation > Crew deployment. Predictive alerting and Admin questions run alongside.

Who owns what

Component	Owner	Its job in one line
Intake bot	Sankalp	Turns the message into a request type and location
Emergency check	Sankalp	Stops 911 situations before anything is stored
Agency and skill routing	Mohammad	Picks the agency and the crew skill
Duplicate detection	Nabeel	Groups reports of the same problem
External factors	Mohammad	Nearby schools and hospitals, and the weather
Credibility and escalation	Sankalp	Reporter track record, and repeat problems
Fake request detection	Nabeel	Risk that nothing will be found
Resolution time	Mohammad	Expected time to fix, as a range
Urgency	Khalid	Critical, High, Medium or Low, with reasons
Prioritisation	Khalid	Ranks the queue
Crew deployment and routing	Mohammad	Assigns crews and plans routes
Predictive alerting	Sankalp	Warns of busy days ahead
Admin questions	Nabeel	Answers staff questions in plain English

The 34 request types

The intake bot can only output one of these codes, or one of the three labels below. Agency, crew skill, default urgency and time on site are looked up from the code rather than guessed. Source: `data/reference/request_catalog.csv`.

Code	Problem	Agency	Crew skill	Urgency	Min
RT001	Pothole	DOT	ROAD_REPAIR	MEDIUM	60
RT002	Street cave-in or sinkhole	DOT	ROAD_REPAIR	HIGH	120
RT003	Other street surface defect	DOT	ROAD_REPAIR	LOW	60
RT004	Street light out or dim	DOT	STREET_LIGHTING	MEDIUM	45
RT005	Traffic or pedestrian signal not working	DOT	TRAFFIC_SIGNALS	HIGH	60
RT006	Broken sidewalk or curb	DOT	SIDEWALK_REPAIR	LOW	90
RT020	Water main break or street water leak	DEP	WATER_MAINS	HIGH	120
RT021	Catch basin clogged or flooding	DEP	SEWER_DRAINAGE	MEDIUM	45
RT022	Street flooding	DEP	SEWER_DRAINAGE	HIGH	60
RT023	Sewer backup or manhole overflow	DEP	SEWER_DRAINAGE	HIGH	90
RT024	Fire hydrant running or damaged	DEP	WATER_MAINS	MEDIUM	45
RT025	Dirty water or low water pressure	DEP	WATER_MAINS	MEDIUM	45
RT040	Missed garbage or recycling collection	DSNY	SANITATION_COLLECTION	LOW	20
RT041	Illegal dumping	DSNY	STREET_CLEANING	MEDIUM	40
RT042	Dirty street or litter	DSNY	STREET_CLEANING	LOW	30
RT043	Derelict vehicle	DSNY	VEHICLE_REMOVAL	LOW	60
RT044	Snow or ice not cleared	DSNY	SNOW_ICE	MEDIUM	45
RT045	Dead animal on street	DSNY	STREET_CLEANING	LOW	20
RT060	Fallen tree or branch	DPR	FORESTRY	MEDIUM	90
RT061	Overgrown tree or branches on wires	DPR	FORESTRY	MEDIUM	60
RT062	Dead or dying tree	DPR	FORESTRY	LOW	60
RT080	No heat or hot water	HPD	HOUSING_INSPECTION	HIGH	45
RT081	Water leak inside building	HPD	HOUSING_INSPECTION	MEDIUM	45
RT082	Plumbing problem in apartment	HPD	HOUSING_INSPECTION	MEDIUM	45
RT083	Pests, mold or unsanitary condition	HPD	HOUSING_INSPECTION	LOW	45
RT084	Electrical problem in building	HPD	HOUSING_INSPECTION	MEDIUM	45
RT100	Rats or rodents	DOHMH	PEST_CONTROL	LOW	40
RT120	Noise from a residence	NYPD	NOISE_PATROL	LOW	20
RT121	Noise from street, business or vehicle	NYPD	NOISE_PATROL	LOW	20
RT122	Blocked driveway	NYPD	PARKING_ENFORCEMENT	MEDIUM	20
RT123	Illegal parking or blocked hydrant	NYPD	PARKING_ENFORCEMENT	LOW	20
RT124	Abandoned vehicle	NYPD	VEHICLE_REMOVAL	LOW	30
RT140	Unsafe building or illegal construction	DOB	BUILDING_INSPECTION	MEDIUM	60
RT141	Elevator not working	DOB	BUILDING_INSPECTION	MEDIUM	45

Label	Meaning	What happens
EMERGENCY	Someone may be in danger	The citizen is told to call 911. Nothing is filed
OTHER	A real question no crew can fix	Goes to a person rather than a crew
SMALLTALK	Hello, thanks	No request is created

Shared values and rules

Allowed values

Value	Allowed values, exactly as written	Where it comes from
Request type code	RT001 to RT141, as listed on page 2	Request catalog
Urgency level	CRITICAL, HIGH, MEDIUM, LOW, always in capitals	Agreed 14 October
Agency	DEP, DOB, DOHMH, DOT, DPR, DSNY, HPD, NYPD	Request catalog
Crew skill	BUILDING_INSPECTION, FORESTRY, HOUSING_INSPECTION, NOISE_PATROL, PARKING_ENFORCEMENT, PEST_CONTROL, ROAD_REPAIR, SANITATION_COLLECTION, SEWER_DRAINAGE, SIDEWALK_REPAIR, SNOW_ICE, STREET_CLEANING, STREET_LIGHTING, TRAFFIC_SIGNALS, VEHICLE_REMOVAL, WATER_MAINS	Request catalog
Borough	BRONX, BROOKLYN, MANHATTAN, QUEENS, STATEN ISLAND	Cleaned 311 data
Time	New York local time, for example 2026-10-14 09:30. Proposed, see below	Cleaned 311 data
emergency_action	CONTINUE, CONFIRM_WITH_CITIZEN, STOP_CALL_911	Emergency check
Outcome of a closed job	GENUINE, UNFOUNDED, INCONCLUSIVE	Crew closing notes
Credibility score	0 to 100. 50 means as expected	Credibility
Credibility band	NEW, LOW, NEUTRAL, TRUSTED	Credibility
Escalation level	NONE, SUPERVISOR, CENTRAL_AUTHORITY	Credibility
Alert severity	WARNING, WATCH, INFO	Predictive alerting

Shared rules we agreed

These come from the 28 September presentation.

1. The emergency check runs first, before anything is stored, and only `CONTINUE` moves on to the next step.
2. Requests are held for a person rather than dispatched when `needs_human` is true, or when the label is EMERGENCY, OTHER or SMALLTALK.
3. The request type code is the shared key. Agency, skill and time on site are looked up from the catalog with it.
4. Low credibility or fake risk can flag a request for review, but does not lower a hazard or remove a request.
5. Critical requests rank first. Trust and fake risk can reduce a score to no less than 0.6 times its value, and never to zero.
6. If a component fails, it falls back to its safe default, which does not lower anyone's score.
7. Nobody is refused automatically.

Decided on 14 October

- Urgency is written in capitals everywhere: CRITICAL, HIGH, MEDIUM, LOW.
- Nabeel owns Admin questions.
- The profanity filter is removed from scope. Messages go straight from the intake bot to the emergency check.

Proposed, for the group to confirm

- All times are New York local time. The processed 311 data already uses it, so nothing needs to change there. A component that expects a timezone could treat a time without one as New York time.

Still open, suggestions welcome

- Who would like to turn an address into map coordinates? The intake bot sends the address as text.
- Crew shifts appear on presentation slide 8, but the crew data no longer has them. Shall we drop them from the slide too?

Khalid

Owns: Urgency, Prioritisation

Receives

From	Field	Meaning
Intake bot	request_type	Request type code, such as RT022
Intake bot	slots.hazard_terms	Hazard words found, such as flooding
Intake bot	slots.time_reference	Time phrase found, such as since this morning
Emergency check	emergency_action	CONTINUE, CONFIRM_WITH_CITIZEN, STOP_CALL_911
Credibility	credibility_score	0 to 100, where 50 means as expected
Credibility	escalation_level	NONE, SUPERVISOR, CENTRAL_AUTHORITY
External factors	to be named by Mohammad	Nearby sensitive sites, weather
Duplicate detection	to be named by Nabeel	Number of people reporting it
Fake request detection	to be named by Nabeel	Risk that nothing is found
Resolution time	to be named by Mohammad	Expected time to fix

Hands on

To	Field	Meaning
Prioritisation	to be named by Khalid	Urgency level and its reasons
Crew deployment	to be named by Khalid	The ranked queue

Points to discuss together

- The group agreed to write urgency in capitals: CRITICAL, HIGH, MEDIUM, LOW. Urgency currently writes Medium, so this may need a small change.
- Could Prioritisation read the escalation level that Credibility sends? Its allowed values are on page 3.
- Could Prioritisation also use the number of duplicate reports, once Nabeel has named that field?
- Could Prioritisation treat a time without a timezone as New York time, so it runs on the 311 data as it is?

Mohammad

Owns: Agency and skill routing, External factors, Resolution time, Crew deployment and routing

Receives

From	Field	Meaning
Intake bot	request_type	Request type code. The key to the catalog
Intake bot	ready_to_submit	Dispatched only when true
Intake bot	needs_human	Held for a person when true
Intake bot	slots.incident_address, slots.cross_street	Location as text, not coordinates
Intake bot	slots.borough	BRONX, BROOKLYN, MANHATTAN, QUEENS, STATEN ISLAND
Emergency check	emergency_action	Only CONTINUE moves on to dispatch
Request catalog	agency, required_skill, est_service_minutes	Looked up by the request type code
Crew table	crew_id, agency, borough, skills, depot_lat, depot_lon, max_jobs, avg_speed_kmh	One row per crew
Prioritisation	to be named by Khalid	The ranked queue

Hands on

To	Field	Meaning
Urgency, Prioritisation	to be named by Mohammad	External factors
Citizen, Urgency	to be named by Mohammad	Expected time to fix, as a range
Credibility	outcome, closed_at, request_type	One row per closed job, outcome one of GENUINE, UNFOUNDED, INCONCLUSIVE
Crew portal	to be named by Mohammad	Each crew's route

Points to discuss together

- Who would like to turn the address into latitude and longitude?
- A job sent to the wrong agency closes as REFERRED and is rerouted. A suggestion: it does not count for or against the reporter.
- Crew shifts: slide 8 lists them, but the crew data does not. Shall we drop them from the slide?
- Already in place: the crew table matches the catalog, with the same 8 agencies and 16 skills.

Nabeel

Owns: Duplicate detection, Fake request detection, Admin questions

Receives

From	Field	Meaning
Intake bot	request_type	Request type code
Intake bot	slots.incident_address, slots.borough	Location as text
Crew deployment	to be named by Mohammad	Closed jobs with their outcome
Credibility	credibility_score	0 to 100, where 50 means as expected

Hands on

To	Field	Meaning
Urgency, Prioritisation	to be named by Nabeel	Number of people reporting it
Credibility	closures_same_place_and_type	Earlier request ids closed at the same place and type in the last 90 days
Urgency, Prioritisation	to be named by Nabeel	Fake risk

Points to discuss together

- What format would suit the duplicate result, so Urgency and Prioritisation can read it easily?
- Could Duplicate detection send the list of earlier closures at the same place? Credibility uses it to escalate at 3 and 5 repeats.
- Admin questions: which stored fields would it like to read? Using the names in this document will keep its answers in line with what the components produced.

Sankalp

Owns: Intake bot, Emergency check, Credibility and escalation, Predictive alerting

Receives

From	Field	Meaning
Citizen	text, citizen_id, borough_hint, history	Only the message text is required
Crew deployment	outcome, closed_at, request_type	Each citizen's past closed jobs
Duplicate detection	closures_same_place_and_type	Repeats at the same place, last 90 days
Weather forecast	wx_precip_mm, wx_snow_cm, wx_tmax_c, wx_tmin_c, wx_gust_kmh	Daily, per borough

Hands on

To	Field	Meaning
Everyone after intake	request_type, confidence, ready_to_submit, needs_human	What the request is, and whether it may go on
Everyone after intake	slots.incident_address, slots.cross_street, slots.borough, slots.landmark	Where it is, as text
Everyone after intake	is_emergency, emergency_action, emergency_category	Emergency result
Urgency, Prioritisation	credibility_score, credibility_band, escalation_level	Track record and repeat problems
Admin portal	borough, complaint_family, lift, severity	Busy day warnings

Points to discuss together

- The intake bot sends the location as text, without map coordinates.
- The credibility score is sent on a 0 to 100 scale, where 50 means as expected.